

Deep Learning for teens

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Transformers

- ▶ Bert are based on Transformer architecture .

Why we need two URLs for BERT

- ▶ **1) Preprocess URL → turns raw text into BERT-ready input**
 - ▶ This module does all the “text cleaning + tokenizing” work
- ▶ **2) Encoder URL → the actual BERT model**
 - ▶ This is the real BERT neural network

TensorFlow Hub

- ▶ TensorFlow Hub is basically a big online library of pretrained machine-learning models that you can download and use right away.

Word embedding vs Bert

▶ BERT

- ▶ is a pretrained model.
- ▶ Is trained before you ever use it, on a massive amount of text (like Wikipedia and books).
- ▶ During this pretraining, it learns:
 - ▶ how sentences are structured
 - ▶ how words relate to each other
 - ▶ how to understand context
- ▶ After that, you can fine-tune it for a specific task, like:
 - ▶ sentiment analysis
 - ▶ question answering
 - ▶ text classification
 - ▶ named entity recognition

Why ordinary word embedding not working when you have a sentence

- ▶ ☆ Why ordinary word embeddings don't work well for sentences
 - ▶ One meaning only
 - ▶ Each word gets one fixed vector, so it can't change meaning based on the sentence.
 - ▶ No sense of order
 - ▶ They don't know which word comes first or last, so the structure of the sentence is lost.
 - ▶ They forget long sentences
 - ▶ They can't keep track of how early words relate to later words.

BERT

- ▶ What is BERT
 - ▶ Think of BERT as a model that **reads text like a human** — not just left-to-right, not just right-to-left, but **both directions at the same time**.
- ▶ BERT can make contextualized embedding means
 - ▶ A contextualized embedding is a word vector whose meaning changes depending on the sentence around it.
 - ▶ In older models (Word2Vec, GloVe), each word had one fixed vector, no matter where it appeared.

[CLS] and [SEP]?

- ▶ BERT doesn't take raw text.
- ▶ It needs special tokens added to the beginning and end of the input so it knows how to process the sentence.
- ▶ [CLS] – Classification token
 - ▶ Added at the very beginning of every input sentence.
 - ▶ BERT uses the embedding of this token as the summary of the entire sentence.
 - ▶ When you do sentiment analysis, spam detection, etc., the model uses the [CLS] vector.
 - ▶ Think of it as:
 - ▶ “This token represents the whole meaning of the sentence.”

[CLS] and [SEP]?

- ▶ **[SEP] – Separator token**
 - ▶ Added **at the end** of every sentence.
 - ▶ Used to separate:
 - ▶ sentence A from sentence B (for tasks like question answering)
 - ▶ or just to mark the end of a single sentence
 - ▶ Think of it as:
 - ▶ **“This marks where the sentence ends.”**